

and switches guarantee isolation. iSCSI should also be implemented on cabling that is independent of standard LAN traffic, since performance can fall due to bandwidth congestion on shared networks. Both the Internet and FC now offer copper and optical cabling options, allowing network consolidation combined with traffic classification.

Storage area network traffic is typically unencrypted, since physical server-to-storage cabling is normally enclosed within secure data centres. For WAN security, iSCSI and fibre channel over Ethernet (FCoE) can utilise Internet Protocol Security (IPSec), which is a protocol suite for securing IP network traffic. Select networking hardware (preferred NICs, TOEs and HBAs) can provide encryption. iSCSI offers the Challenge-Handshake Authentication Protocol (CHAP) user names and passwords as an authentication mechanism to limit connectivity between chosen initiators and targets.

Until recently, iSCSI was not considered an enterprise-grade storage option, primarily due to the use of slower 100Mbps and 1000Mbps Ethernet, compared to 4Gbps optical infrastructure. With the current 10Gbps or 40Gbps FC, bandwidth availability is now similar for both.

The use of iSCSI extends a SAN beyond the limit of local cabling, facilitating storage consolidation in local or remote data centres. Because iSCSI structures are logical, new storage allocations are made using only software configurations, without the need for additional cable or physical disks. iSCSI also simplifies data replication, migration and disaster recovery using multiple remote data centres.

iSCSI fundamentals

The iSCSI protocol functions in a familiar client-server configuration. Client systems configure initiator software to send SCSI commands to remote server storage targets. The accessed iSCSI target appears on the client system as local, unformatted SCSI block devices, identical to devices connected with SCSI cabling, FC direct attached, or FC switched fabric.

iSCSI component terminology

1. **Initiator:** This is an iSCSI client, typically available as software but also implemented as iSCSI HBAs. Initiators must be given unique names.
2. **Target:** This is an iSCSI storage resource, configured for connection from an iSCSI server. Targets must be given unique names. A target provides one or more numbered block devices called logical units. An iSCSI server can provide many targets concurrently.
3. **ACL:** An Access Control List (entry) is an access restriction using the node IQN (commonly the iSCSI initiator name) to validate access permissions for an initiator.
4. **Discovery:** This involves querying a target server to list configured targets. Using targets requires additional

access steps.

5. **IQN:** An iSCSI Qualified Name, this is a worldwide unique name used to identify both initiators and targets, in the following mandated naming format:

`Inqn.YYYY-MM.com.reversed.domain[:optional_string]`

IQN signifies that this name will use a domain as its identifier. YYYY-MM is the first month in which the domain name was owned. *com.reversed.domain* is the reversed domain name of the organisation creating the iSCSI name. *optional_string* is an optional, colon-prefixed string assigned by the domain owner as desired while remaining unique, worldwide. It may include colons to separate the organisation boundaries.

6. **Login:** This authenticates to a target or to a LUN to begin the client's block device use.
7. **LUN:** A Logical Unit Number, it numbers block devices attached to and available through a target. One or more LUNs may be attached to a single target although, typically, a target provides only one LUN.
8. **Node:** This is an iSCSI initiator or iSCSI target identified by its IQN.
9. **Portal:** This is an IP address and port on a target or initiator used to establish connections. Some iSCSI implementations use the terms portal and node interchangeably.
10. **TPG or Target Portal Group:** This is the set of interface IP addresses and TCP ports to which a specific iSCSI target will listen. Target configuration can be added to TPG to coordinate settings for multiple LUNs.

iSCSI uses ACLs to perform LUN masking, managing the accessibility of appropriate targets and LUNs to initiators. Access to targets may also be limited with CHAP authentication. iSCSI ACLs are similar to FC's use of a device's worldwide numbers (WWNs) for soft zoning management restrictions, although FC switch level compulsory port restrictions (hard zoning) have no comparable iSCSI mechanisms. Ethernet VLANs could provide similar isolation security.

Unlike local block devices, iSCSI network-accessed block devices are discoverable from many remote initiators. Typical local file systems (ext4, XFS) do not support concurrent multi-system mounting, which can result in significant file system corruption. Clustered file systems resolve multiple-system access by the use of the Global File System (GFS2), which is designed to provide distributed file locking and concurrent multi-node file system mounting. An attached iSCSI block device appears as a local SCSI block device (sdx) for use underneath a local file system, swap space or raw database installation.

An overview of the iSCSI target

In original SCSI protocol terminology, a target is a single